



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to DEVA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

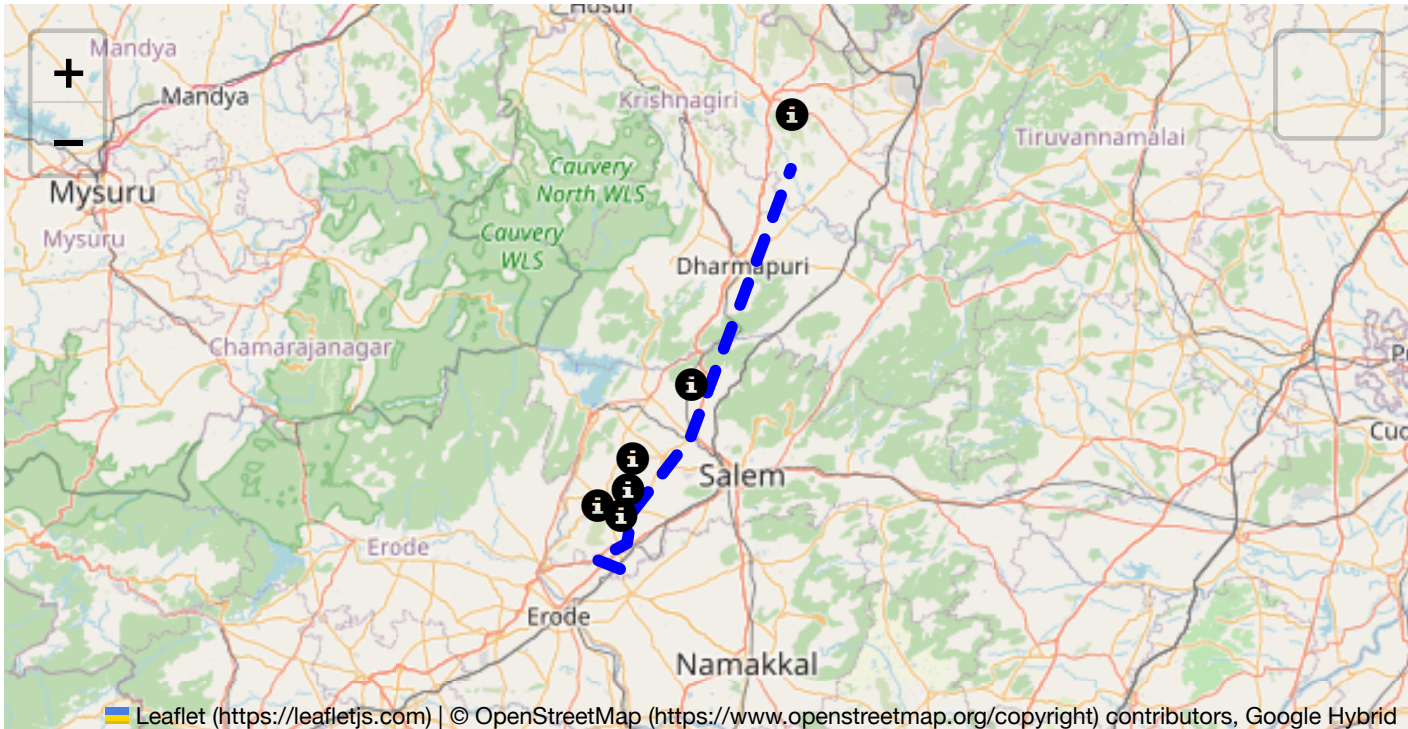
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

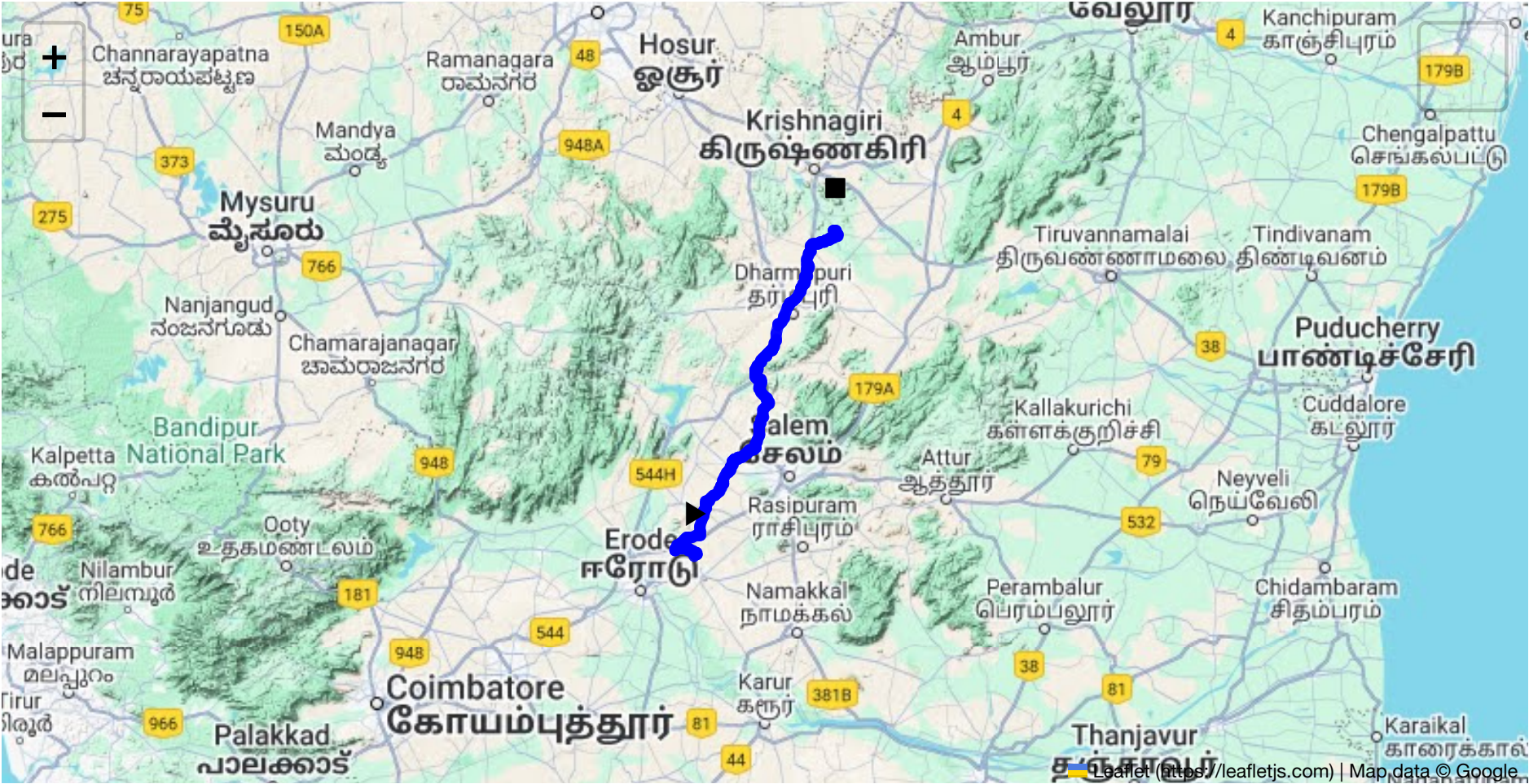
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 138.08 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours

Start: (11.4381, 77.8734)
End: (12.3582266, 78.2742459)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The described route spans approximately 138.08 kilometers starting at CVQF+23W in Sangagiri, traversing multiple towns and regions, and concluding at 975F+RJ4 in Maruderi. Key stopovers include locations in Bhavani Main Road, Padaiveedu, Ayveli, Erumaipatti, and Salem. The journey involves navigating a mix of state highways (SH 86) and regional roads, commonly used by long-haul trucks.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate, with a monsoon season from October to December. This can cause heavy rainfall, particularly affecting the Salem region, which might result in waterlogged roads, reduced visibility, and increased skidding risk. Summer months (March to May) could see extreme temperatures leading to increased vehicle overheating risks.

3. Analysis of Traffic Patterns

- **Sangagiri to Salem:** Expect moderate congestion, especially around Salem city during peak hours (8-10 AM and 5-8 PM).
- **Salem to Maruderi:** Traffic tends to be lighter, but agricultural vehicles and local commuters can slow progress towards smaller towns.
- **Congestion-Prone Areas:** Salem city center and key intersections on Bhavani Main Road can experience significant delays.

4. Assessment of Road Quality and Infrastructure

- **Road Conditions:** State highways generally have better maintenance than local roads, which may suffer from potholes and uneven surfaces, particularly post-monsoon.
- **Bridge and Overpass Checks:** Ensure weight limits are adhered to, especially on older bridges along the SH 86.

5. Suggestions for Alternative Routes for Emergencies

Where local routing becomes difficult, consider:

- **Alternate via SH 79:** Divert through NH 544 and NH 79 towards Salem for better road quality.
- **Substitute through SRV Main Road:** Avoid heavily trafficked areas by utilizing smaller by-roads, though check for width and clearance.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is subject to strict regulations in Tamil Nadu, including specific banning from certain areas and time-bound restrictions, particularly in urban centers like Salem. Ensure proper permits and adherence to designated travel times.

7. Overview of Historical Incidents

Incidents involving heavy vehicles mainly occur near congested urban areas. Notably, Salem has seen incidents attributed to poor road discipline and occasional mechanical failures. Remain vigilant in urban traffic and during rain.

8. Environmental Considerations and Sensitive Areas

The route passes through several agricultural regions. Caution should be exercised to prevent pollution from hazardous materials, given proximity to crops and water sources.

9. Analysis of Communication Coverage

While Salem and nearby towns generally have good cellular service, more remote areas such as rural stretches near Erumaipatti might experience occasional dead zones.

10. Estimated Emergency Response Times

- **Sangagiri to Salem:** Approximately 30-45 minutes response time due to proximity to emergency services.
- **Salem and Beyond:** Can extend to 1 - 1.5 hours in rural stretches, especially during peak times.

12. Overall Summary of Risk Assessment

The route presents moderate risk primarily due to weather-related hazards and urban congestion. While the infrastructure supports heavy vehicle travel, caution is needed during monsoon seasons and when entering urban centers. Constant vigilance, regular vehicle checks, and adherence to local regulations are essential for maintaining safety during transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	High	11.45542, 77.81725	15 KM/Hr
7	Turn	High	11.45605, 77.81683	15 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	Medium	11.56072, 77.89919	30 KM/Hr
11	Turn	Medium	11.56140, 77.89991	30 KM/Hr
12	Turn	Medium	11.59085, 77.90592	30 KM/Hr
13	Turn	Medium	11.59253, 77.90690	30 KM/Hr
14	Turn	Medium	11.63848, 77.94825	30 KM/Hr
15	Turn	High	11.64197, 77.95197	15 KM/Hr
16	Turn	Medium	11.64558, 77.95108	30 KM/Hr
17	Turn	Medium	11.67883, 77.96865	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71386, 77.99552	30 KM/Hr
20	Turn	Medium	11.71735, 78.00019	30 KM/Hr
21	Turn	Medium	11.72765, 78.02291	30 KM/Hr
22	Turn	High	11.73785, 78.03496	15 KM/Hr
23	Turn	High	11.73653, 78.03886	15 KM/Hr
24	Turn	Medium	11.74252, 78.04705	30 KM/Hr
25	Turn	Medium	11.77998, 78.05786	30 KM/Hr
26	Turn	Medium	11.86915, 78.08382	30 KM/Hr
27	Turn	Medium	11.87998, 78.08170	30 KM/Hr
28	Turn	Medium	11.91266, 78.05852	30 KM/Hr
29	Turn	Medium	11.93746, 78.05331	30 KM/Hr
30	Turn	Medium	11.95531, 78.04698	30 KM/Hr
31	Turn	High	11.95854, 78.04990	15 KM/Hr
32	Turn	High	11.96015, 78.04945	15 KM/Hr
33	Turn	Medium	11.96370, 78.05142	30 KM/Hr
34	Turn	Medium	11.96433, 78.05113	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
35	Turn	High	11.96542, 78.04938	15 KM/Hr
36	Turn	High	11.96650, 78.04927	15 KM/Hr
37	Turn	Medium	12.17041, 78.16823	30 KM/Hr
38	Turn	Medium	12.22515, 78.19158	30 KM/Hr
39	Turn	Medium	12.22995, 78.19466	30 KM/Hr
40	Turn	Medium	12.24547, 78.19501	30 KM/Hr
41	Turn	Medium	12.24685, 78.19561	30 KM/Hr
42	Turn	Medium	12.25120, 78.20010	30 KM/Hr
43	Turn	High	12.31455, 78.20749	15 KM/Hr
44	Turn	Medium	12.31797, 78.21478	30 KM/Hr
45	Turn	High	12.31905, 78.21568	15 KM/Hr
46	Turn	Medium	12.32419, 78.23951	30 KM/Hr
47	Turn	High	12.32373, 78.24196	15 KM/Hr
48	Turn	Medium	12.32545, 78.24407	30 KM/Hr
49	Turn	Medium	12.32938, 78.25293	30 KM/Hr
50	Turn	Medium	12.33767, 78.27118	30 KM/Hr
51	Turn	Medium	12.34242, 78.28153	30 KM/Hr
52	Turn	High	12.34515, 78.28243	15 KM/Hr
53	Turn	Medium	12.35215, 78.27693	30 KM/Hr
54	Turn	High	12.35226, 78.27648	15 KM/Hr
55	Turn	High	12.35656, 78.27468	15 KM/Hr
56	Turn	Medium	12.35747, 78.27478	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
32	clinic	G.K Clinic - Beruhalli	12.3566347, 78.2746656	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
31	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56072, 77.89919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56140, 77.89991



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59085, 77.90592



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59253, 77.90690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63848, 77.94825



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64558, 77.95108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67883, 77.96865



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17041, 78.16823



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22515, 78.19158



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22995, 78.19466



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24547, 78.19501



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24685, 78.19561



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25120, 78.20010



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31455, 78.20749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31797, 78.21478



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31905, 78.21568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32419, 78.23951



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.32373, 78.24196



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35215, 78.27693



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35226, 78.27648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35656, 78.27468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35747, 78.27478

